

Trainings ▾

Preparatory Steps

Remove or disconnect the following:

- Charge air cooler connection hose and hose clamp from the Exhaust Gas Recirculation (EGR) mixer inlet. Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/35/35-010-027.html>)
- Air compressor supply line. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html>)
- V-band clamp between the EGR mixer and the charge air cooler inlet adapter.
- Wiring harness.
- Loosen the hose clamps at the EGR return tube hose. Refer to Procedure 011-025 (</qs3/pubsys2/xml/en/procedures/35/35-011-025.html>)



Remove

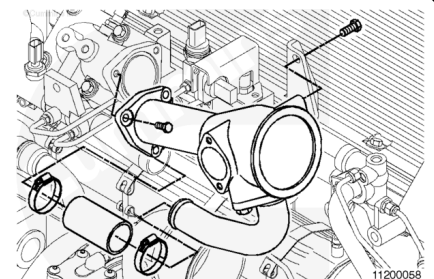
Remove the capscrews at the three bolt flange between the EGR mixer and the air intake horn.

Remove the capscrews at the EGR mixer support bracket.

Remove the EGR mixer.

Remove the mixer support bracket.

Remove the EGR return tube hose.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

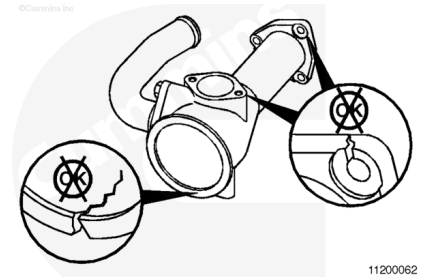
Use compressed air to blow out any debris or soot buildup inside the EGR mixer. A thin layer of soot is normal.

Clean all of the mounting surfaces with solvent.

Dry with compressed air.

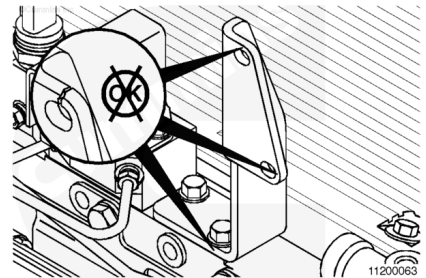
Inspect the EGR mixer for cracks.

Replace the EGR mixer if cracks are found.



Inspect the EGR mixer bracket for cracks.

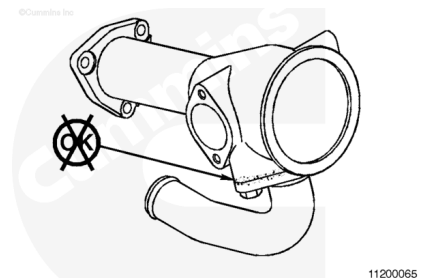
Replace the EGR mixer bracket if cracks are found.



Inspect the EGR inlet pipe, that leads into the EGR mixer, for cracks or damage.

Replace the EGR inlet pipe if cracks or damage is found.

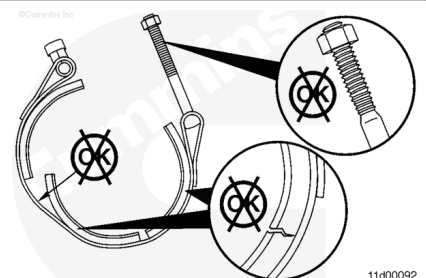
Check for soot near the two-bolt flange at the connector of the EGR inlet pipe into the EGR mixer. If found, replace the o-ring.



Inspect the V-band clamps for signs of overextension. The band **must not** be bent or cracked.

Inspect the V-band clamp threads for damage.

Replace the V-band clamp if damage is found.

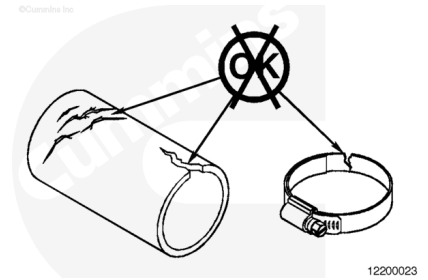


Inspect the hose clamp(s) for cracks or other damage.

Replace hose clamp(s) if cracks or other damage is found.

Inspect hose(s) for cracks or other damage.

Replace hose(s) if cracks or other damage is found.



Install

Install the hose at the EGR mixer inlet.

Install a new o-ring between the EGR mixer and the air intake connection.

Install the EGR mixer.

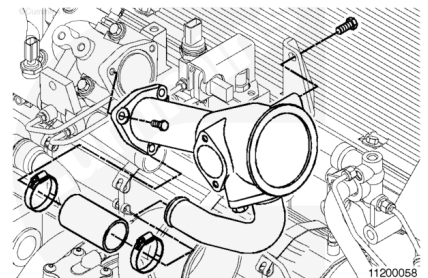
Install and tighten the capscrews.

Torque Value: 20 n•m [15 ft-lb]

Install the EGR mixer bracket. Loosely install the four capscrews.

Tighten capscrews in a criss-cross fashion.

Torque Value: 20 n•m [15 ft-lb]



Finishing Steps

Install or connect the following:

- V-band clamp with a new o-ring on the inlet adapter and the EGR mixer.
- Air compressor supply line. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html>)
- Charge air cooler connection. Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/35/35-010-027.html>)
- Wiring harness.
- Hose and clamps at EGR return connection. Refer to Procedure 011-025 (</qs3/pubsys2/xml/en/procedures/35/35-011-025.html>)



Start and operate the engine and check for leaks at the air connections.

If malfunction resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment **must** be inspected. Refer to Procedure 014-013 (</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>)

Last Modified: 07-Oct-2011
